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Starting the proof of

toric mirror symmetry

What we've done so far

- > Background on toric geometry
- > Whirlwind tour of ∞ -categories
- > (Commutative) monoids & Spectra
- > The functor of points
- > Set up basic Spectral algebraic geometry
 - We can then immediately formulate toric geometry over the universal base, the Sphere Spectrum.

Rest of the semester Proving toric mirror symmetry. Let's recall the most basic statement.

Nth. Given a fan (N, Σ) , we write $X_\sigma = \text{Spec}(S\langle M\sigma^\vee \rangle)$ and $X_\Sigma = \text{colim}_{\sigma \in \Sigma^{\text{op}}} X_\sigma$ for the toric stack over the sphere.

- > Nothing specific about the base will be used, so if it makes it mentally easier, feel free to replace S by $\mathbb{Z}$ or a field k .
 - And $\text{Sp} = \text{Mod}_S$ by $\text{Mod}_{\mathbb{Z}}$ or Mod_k .

Recall A fan (N, Σ) is

- (1) Smooth if each cone $\sigma \in \Sigma$ can be generated by part of a basis for N .
- (2) projective if there is a rational polytope $P \subset M_{\mathbb{R}}$ such that of full dimension such that Σ is the normal fan Σ_P of P .

Thm (FLTZ) Let (N, Σ) be a smooth projective fan. There is a fully faithful symmetric monoidal functor

$$K: (\mathrm{QCoh}(X_{\Sigma}/T), \otimes) \hookrightarrow (\mathrm{Sh}(M_{\mathbb{R}}, Sp), *)$$

↙ Convolution

The image of K consists of those sheaves with microsupport in the $\mathbb{R}_{>0}$ -conic Lagrangian Λ_{Σ} in

$$T^*M_{\mathbb{R}} \simeq M_{\mathbb{R}} \times N_{\mathbb{R}}$$

defined by

$$\Lambda_{\Sigma} := \bigcup_{m \in M, \sigma \in \Sigma} (m + \sigma^{\vee}) \times \sigma.$$

Proof Overview

Strategy to construct κ .

(1) We start with an affine version of the problem. For $\sigma \in \Sigma$, we want a combinatorial description of $\mathrm{QCoh}(X_\sigma/T)$.

- we'll define a poset $\Theta(\sigma)$ of some specific closed polyhedral subsets of $M_{\mathbb{R}}$. This will be a symmetric monoidal category under Minkowski sum

$$S+T = \{s+t \mid s \in S \text{ and } t \in T\}$$

- we'll construct a symmetric monoidal equivalence

$$(\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \xrightarrow{\sim} (\mathrm{QCoh}(X_\sigma/T), \otimes)$$

- we'll construct a fully faithful lax sym. monoidal structure

$$(\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \hookrightarrow (\mathrm{Sh}(M_{\mathbb{R}}; \mathrm{Sp}), *).$$

(2) We'll show that all the functors in (1) are compatible with inclusions of cones.

- By Zariski descent, taking the limit over $\sigma \in \Sigma^{\text{op}}$, we obtain the functor K .

$$\text{QCoh}(X_{\Sigma}/T) \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \text{QCoh}(X_{\sigma}/T) \hookrightarrow \text{Sh}(M_{\mathbb{R}}, \text{Sp})$$

- This will involve an analogous gluing formula for $\text{Sh}(M_{\mathbb{R}}, \text{Sp})$ in terms of subcategories of modules over idempotent algebras.

Note. To construct these functors, we'll use the six-functor formalism on topological spaces. For example, when $N = \mathbb{Z}$ and $\sigma = \mathbb{R}_{>0}$, we'll have:

> $\Theta(\sigma) = (\mathbb{Z}, \leq)^{\text{op}}$ with sym. mon. structure addition

> The functor $\text{Fun}(\mathbb{Z}, \text{Sp}) \longrightarrow \text{Sh}(\mathbb{R}, \text{Sp})$ will send $n \in \mathbb{Z}$ to the dualizing sheaf $\omega_{(-\infty, n)} = p^!(S)$ $p: (-\infty, n) \rightarrow *$

> The equivalence $\text{Fun}(\mathbb{Z}, \text{Sp}) \xrightarrow{\sim} \text{QCoh}(A^1/G_m)$ is Simpson's

equivalence that classifies the universal line bundle $\mathcal{O}(1)$ with a section $\cdot x: \mathcal{O} \rightarrow \mathcal{O}(1)$

Strategy to characterize the image of k

> The idea is that the relationship to the combinatorial description of quasicoherent sheaves comes from the theory of exit-path ∞ -categories:

- If (X, P) is any reasonable stratified space, there is an ∞ -category $\text{Exit}(X, P)$ and an equivalence

$$\text{Fun}(\text{Exit}(X, P), \text{An}) \underset{\text{or Sp}}{\simeq} \left(\begin{array}{l} \text{Sheaves on } X \text{ that} \\ \text{are locally constant} \\ \text{along the stratification} \end{array} \right)$$

'constructible sheaves'

- Moreover, this ∞ -category is often a 1-category.

> The theory of sheaves with specified microsupport is a generalization of the theory of constructible sheaves, but in the context of manifolds.

- Developed by Kashiwara-Schapira
- Input: (real analytic) manifold X , $\Lambda \subset T^*X$ closed, $\mathbb{R}_{\geq 0}$ -conic (subanalytic) Lagrangian
- Output: A full subcategory $\mathrm{Sh}_\Lambda(X, \mathbb{S}p) \subset \mathrm{Sh}(X, \mathbb{S}p)$ closed under limits and colimits. When Λ is the union of conormals to a Whitney stratification, $\mathrm{Sh}_\Lambda(X, \mathbb{S}p)$ recovers constructible sheaves
- The theory is involved to set up in general, but greatly simplifies when X is a vector space
- Careful analysis is needed to identify $\mathrm{im}(k)$

Conjecture (Nadler, vague formulation). The ∞ -category $\mathrm{Sh}_\Lambda(X, \mathbb{S}p)$ should admit an 'exit-path' description. That is, there should be a geometrically-defined ∞ -category $\mu\mathrm{Exit}(X, \Lambda)$ and an equiv.

$$\mathrm{Sh}_\Lambda(X, \mathbb{S}p) \simeq \mathrm{Fun}^{\mathrm{conditions}}(\mu\mathrm{Exit}(X, \Lambda), \mathbb{S}p).$$

> Peter is very interested in this conjecture!

The Combinatorial model

Def Let N be a lattice. A subset $P \subset M_{\mathbb{R}}$ is polyhedral if P can be written as the Minkowski sum of a polytope and a polyhedral cone. We write $\text{Poly}(M)$ for the poset of polyhedral subsets of $M_{\mathbb{R}}$ ordered by inclusion.

Observations

- (1) Every polyhedral subset is closed.
- (2) The Minkowski sum of two polytopes is a polytope and the sum of two polyhedral cones is a polyhedral cone. So $\text{Poly}(M)$ is closed under Minkowski sum.
- (3) Minkowski sum makes $\text{Poly}(M)$ into a symmetric monoidal category with unit $0 = \{0\}$.
- (4) If $P \subset M_{\mathbb{R}}$ is a polyhedral cone, then $P + P = P$.

Def. Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Write

$$\Theta(\sigma) \subset \text{Poly}(M)$$

for the full subposet

$$\Theta(\sigma) = \{m + \sigma^{\vee} \mid m \in M\}$$

Note Each $\Theta(\sigma)$ is closed under Minkowski sum, but does not usually contain the unit 0. Since $\sigma^{\vee} \subset M_{\mathbb{R}}$ is a cone, $\sigma^{\vee} + \sigma^{\vee} = \sigma^{\vee}$. So we see that $\Theta(\sigma)$ is a symmetric monoidal category under Minkowski sum, but with unit $\sigma^{\vee}$.

Def. Let N be a lattice and $i: \sigma \hookrightarrow \tau$ an inclusion of cones in $N_{\mathbb{R}}$. Write $\Theta(i): \Theta(\tau) \rightarrow \Theta(\sigma)$ for the functor given by

$$m + \tau^{\vee} \longmapsto m + \sigma^{\vee}$$

Note that this functor is symmetric monoidal (since everything is a poset, there are no coherences to check, and this is a property).

The assignment $\sigma \mapsto \Theta(\sigma)$ assembles into a functor

$$\Sigma^{\text{op}} \longrightarrow \text{CMon}(\text{Cat}_{\infty})$$

Digression: idempotent algebras

> we don't have enough time in the remainder of this lecture to construct the functor

$$\mathrm{Fun}(\Theta(\sigma)^q, \mathrm{Sp}) \longrightarrow \mathrm{QCoh}(X_\sigma/T)$$

and prove it is an equivalence. So, instead, we'll explain a concept that will appear later and is related to the definition of $\Theta(\sigma)$.

Def Let $\mathcal{C}$ be a symmetric monoidal ∞ -category. A map $u: \mathbb{1} \rightarrow X$ exhibits X as an idempotent object if

$$\mathrm{id}_X \otimes u: X \simeq X \otimes \mathbb{1} \longrightarrow X \otimes X$$

is an equivalence.

A commutative algebra $A \in \mathrm{CAlg}(\mathcal{C})$ is an idempotent algebra if the multiplication

> These are actually the same notion.

Thm. Let $\mathcal{C}$ be a symmetric monoidal category

(1) If $u: \mathbb{1} \rightarrow X$ is an idempotent object, there is a unique commutative algebra structure on X with unit u .

(2) If A is an idempotent algebra, then the unit $\mathbb{1} \rightarrow A$ makes A into an idempotent object.

(3) The functor

$$\text{CAlg}(\mathcal{C}) \longrightarrow \mathcal{C}_{\mathbb{1}}$$

$$A \longmapsto [\text{unit } \mathbb{1} \rightarrow A]$$

restricts to an equivalence between idempotent algebras and idempotent objects

very important

(4) If A is an idempotent algebra, then the forgetful functor $\text{Mod}_A(\mathcal{C}) \rightarrow \mathcal{C}$ is fully faithful with image those objects $Y \in \mathcal{C}$ such that $\text{id}_Y \otimes \text{unit}: Y \simeq Y \otimes \mathbb{1} \rightarrow Y \otimes A$ is an equiv.

Note Hem (4) says it is a property to be an A -module, not extra structure

Ex If N is a lattice and $\sigma \subset N_{\mathbb{R}}$ is a cone, then $\sigma^\vee$ is an idempotent algebra in $\text{Poly}(M)$. The poset $\Theta(\sigma)$ is a full subcategory of

$$\text{Mod}_{\sigma^\vee}(\text{Poly}(M))$$

Ex Let A be an ordinary ring and $S \subset A$ a subset. Then the localization is an idempotent algebra in $\text{Mod}_A^\heartsuit$

Ex For any prime p , the field $\mathbb{F}_p$ is an idempotent algebra in $\text{Mod}_{\mathbb{Z}}^\heartsuit$

> However, $\mathbb{F}_p$ is not an idempotent algebra in $\text{Mod}_{\mathbb{Z}} \simeq D(\mathbb{Z})$. This is because $\mathbb{F}_p \otimes_{\mathbb{Z}}^L \mathbb{F}_p \not\cong \mathbb{F}_p$. Instead,

$$H_1(\mathbb{F}_p \otimes_{\mathbb{Z}}^L \mathbb{F}_p) \simeq \text{Tor}_1^{\mathbb{Z}}(\mathbb{F}_p, \mathbb{F}_p) \simeq \mathbb{F}_p.$$

Ex. One can show that $\mathbb{Q}$ is an idempotent algebra in S_p , but $\mathbb{F}_p$ is not.

Idea for why $\mathbb{F}_p$ isn't an idempotent algebra. In S_p and $D(\mathbb{Z})$, one needs to remember how $p=0$, not just that $p=0$.

Prop. Let $\mathcal{C}$ be a sym mon ∞ -category. A commutative algebra $A \in \text{CAlg}(\mathcal{C})$ is idempotent if and only if for all $B \in \text{CAlg}(\mathcal{C})$, the anima $\text{Map}_{\text{CAlg}(\mathcal{C})}(A, B)$ is empty or terminal. Moreover, $\text{Map}_{\text{CAlg}(\mathcal{C})}(A, B) \neq \emptyset$ if and only if B is an A -module.

Exercise. If $A, B \in \text{CAlg}(\mathcal{C})$ are idempotent algebras, then $A \otimes B$ is an idempotent algebra.

Upshot. For any sym mon ∞ -category $\mathcal{C}$, the ∞ -category $\text{CAlg}^{\text{idem}}(\mathcal{C})$ of idempotent algebras is a poset with binary meets (given by $\otimes$).

Smashing localizations

Def. A symmetric monoidal functor $L: \mathcal{C} \rightarrow \mathcal{D}$ is a smashing localization if there is an idempotent algebra $A \in \text{Alg}(\mathcal{C})$ such that L factors as

$$\mathcal{C} \xrightarrow{A \otimes (-)} \text{Mod}_A(\mathcal{C}) \xrightarrow{\sim} \mathcal{D}$$

- > So L admits a lax sym monoidal right adjoint F .
- > In this case, A is the unique commutative algebra structure on the idempotent object $F(1_{\mathcal{D}}) \simeq FL(1_{\mathcal{C}})$ with unit the adjunction unit $1_{\mathcal{C}} \rightarrow FL(1_{\mathcal{C}})$.
- This agrees with the commutative algebra structure on $F(1_{\mathcal{D}})$ coming from the fact that F is lax sym monoidal.

Def. Let $L: \mathcal{C} \rightarrow \mathcal{D}$ be a symmetric monoidal left adjoint with right adjoint $F: \mathcal{D} \rightarrow \mathcal{C}$, and let $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Write

$$\alpha_{X,Y}: X \otimes_{\mathcal{C}} F(Y) \rightarrow F(L(X) \otimes_{\mathcal{D}} Y)$$

for the map adjoint to

$$L(X \otimes_E F(Y)) \simeq L(X) \otimes_D LF(Y) \xrightarrow{\text{id} \otimes \text{counit}} L(X) \otimes_D Y.$$

We say that the adjunction $L \dashv F$ satisfies the projection formula if for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, the map $\alpha_{X,Y}$ is an equiv

Recognition Principle for smashing localizations. A sym mon left adjoint $L: \mathcal{C} \rightarrow \mathcal{D}$ is a smashing localization if and only if L admits a fully faithful right adjoint F and the adjunction $L \dashv F$ satisfies the projection formula.